The Effects of Patient and Hospital Factors on Hospital-Acquired Infection Risk.

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2026-10-31

1 Overview

Hospital-acquired infections (HAIs), or healthcare-associated infections, are a major problem in United States hospital systems. Hospital-acquired infections are infections that a patient contracts while staying at the hospital. HAIs are linked to many factors, such as the patient's age or length of stay at a hospital, as well as the the hospital's infrastructure, such as the number of available hospital beds and hospital staff. Knowing how these factors affect a patients' risk of acquiring hospital-acquired infections is important for hospital leaders and systems to implement policies in minimizing the risk of hospital-acquired infections and better patient care.

The main research question we will investigate is: How do patient and hospital factors affect the risk of a hospital-acquired infection in a hospital?

We define what "Patient" and "Hospital" factors are in section two.

2 Data Description and Processing

The dataset contains data collected from hospitals in the United States, with each row representing one hospital. The dataset contains 113 rows (observations) and 11 columns (variables).

We categorized the variables in the data set in terms of "Patient Factors" and "Hospital Factors". "Patient Factors" are variables that reference information about a patient, such as how long they stay at the hospital, and their age. "Hospital Factors" are variables associated with the hospital, such as the number of beds the hospital has and the number of nurses for staff. Additionally, we placed "Culture" and "Xray" variables under the group "Hospital Factors", as these are tests conducted by hospitals on patients.

Patient Factors:

Stay (numeric): average length of patient stay (days)

Age (numeric): average patient age (years)

Hospital Factors:

Culture (numeric): Ratio of number of cultures (testing a patient's sample for infection-causing microorganisms) performed to number of patients without signs or symptoms of hospital-acquired infection, times 100.

Xray (numeric): Ratio of number of X-rays (taking an image of the body, particularly the chest, to look for conditions such as pneumonia) performed to number of patients without signs or symptoms of pneumonia, times 100.

Beds (numeric): Average number of beds in hospital during study period.

MedSchool (categorical - factor): Hospital has an associated medical school, where:
–> 1 = Yes
–> 2 = No

We decided to factor this categorical variable, MedSchool, as there are different "groups" or levels for medical school. For example, "1" represents "Yes" as schools with a medical school, and "2" represents "No" as schools without a medical school.

Region (categorical - factor): Geographic region, where:
–> 1 = NE
–> 2 = NC
–> 3 = S
–> 4 = W

We decided to factor this categorical variable, Region, as there are different "groups" of regions. One group of hospitals are located in the Northeast, and three distinct groups are located in the distinct regions of the North Central, South, and West in the United States.

Census (numeric): Average number of patients in hospital per day during study period.

Nurses (numeric): Average number of full-time equivalent registered and licensed practical nurses during study period (number full-time plus one half the number part time).

Facilities (numeric): Percent of 35 potential facilities and services that are provided by the hospital.

Outcome/ Dependent Variable: Our dependent variable is infection risk (named InfctRsk in our dataset). InfctRsk (numeric): Average estimated probability of acquiring infection in hospital (percent)

The data set was checked for missing values. Of the original 116 observations, we removed 3 observations that contained missing values leaving 113 complete observations.

We used the InfctRsk (infection risk) variable as our outcome.

We did not find or remove any outliers in our dataset.

3 Data Exploration

Our first variable to explore is our outcome variable infection risk, labeled InfctRsk in our dataset. This numeric variable is the average estimated probability of acquiring infection in a hospital, written as a percent.

According to the descriptive statistics table "Infection Risk (InfctRsk) Descriptive Statistics", the minimum average infection risk is 1.3% and the average maximum infection risk is 7.8%. The data centers around a mean of 4.35% and the median average infection risk is 4.4%.

Table 1: Infection Risk (InfctRsk) Descriptive Statistics

Min	Mean	Median	Max	StdDev
1.3	4.35	4.4	7.8	1.34

To further explore the infection risk (InfctRsk) variable, a histogram showing the distribution of the variable is constructed. This displays an approximately normal distribution, with one peak.

The density curve fits the data pretty well.

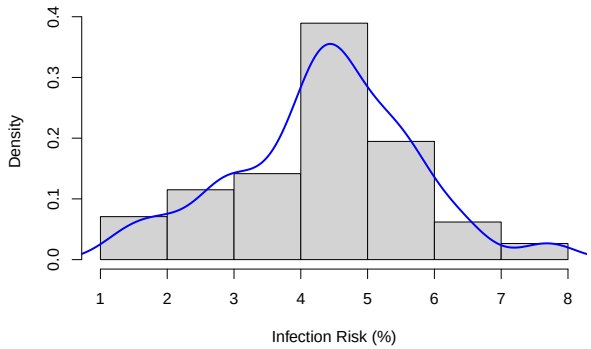


Figure 1: Histogram showing distribution of InfctRsk Variable, with Density Curve Overlaid

The descriptive statistics table matches up with the histogram as we can see that there is a minimum somewhere between 1% and 2% (Actual minimum: 1.3%), a maximum is located between 7% and 8% (actual: 7.8%), and the average/mean for the InfctRsk variable is between 4% and 5% (Actual average or mean: 4.35%).

3.1 Patient Factors

The two "Patient" factors in our dataset are Stay and Age. We explored the distributions of these variables below in sections 3.1.1 and 3.1.2.

3.1.1 The Variable Stay

We start with the variable Stay, which measures average length of patient stay, in days, at a hospital.

The table "Stay Descriptive Statistics" for the variable Stay tell us that the minimum average length that patients stayed at a hospital is 6.7 days, and the maximum average length that patients stayed at a hospital is 19.56 days. The largest cluster of hospitals have patients stay for an average length of 9.65 days.

Table 2: Stay Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
6.7	9.65	9.42	19.56	1.91

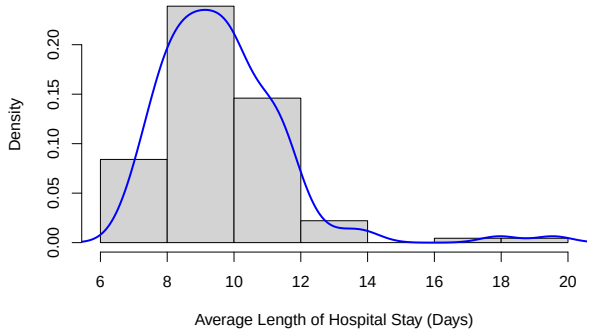


Figure 2: Histogram showing distribution of Stay Variable, with Density Curve Overlaid

The histogram's distribution is skewed to the right, with the average length of Stay across most hospitals is less than two weeks. The largest cluster of hospitals maintains an average patient stay of 8 to 10 days. Based on the skewness and outliers, we can see that there are few hospitals that have much longer average stays, ranging from 16 to 20 days.

The density curve fits the histogram's distribution pretty well.

3.1.2 The Variable Age

The variable Age measures the average patient age is in years.

The histogram for the variable Age reveals an approximately normal distribution, with one peak between 50 and 55 years. This shape is supported by the corresponding descriptive statistics table "Age Descriptive Statistics", which prove that the mean average patient age is 53.23 years old. The minimum average patient age is in the range of 35 and 40 years, and the corresponding descriptive statistics reveal an actual minimum of 38.8 years. Additionally, the histogram reveals a maximum average age at one hospital is in the range of 65 and 70 years, with the descriptive statistics revealing the actual maximum average age to be 65.9 years old

The density curve fits the histogram's distribution well, confirming a normal distribution of the variable Age.

Table 3: Age Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
38.8	53.23	53.2	65.9	4.46

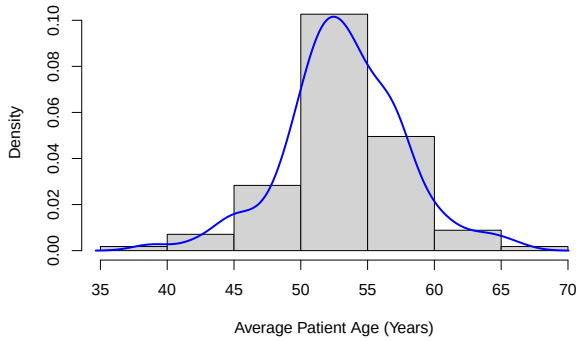


Figure 3: Histogram showing distribution of Age Variable, with Density Curve Overlaid

3.2 Hospital Factors

There are eight "Hospital" factors in our dataset: Culture, Xray, Beds, MedSchool, Region, Census, Nurses, and Facilities.

We explore each variable in the following sections.

3.2.1 Culture

The variable Culture measures the ratio of number of cultures (testing a patient's sample for infection-causing microorganisms) performed to number of patients without signs or symptoms of hospital-acquired infection, times 100. For example, if we use the average ratio of number of cultures from this variable of 15.79, that means that for every 100 hospital patients without signs or symptoms of hospital-acquired infection, 15.79 cultures were performed.

The histogram distribution is skewed to the right, with an average between 10 and 20. The descriptive statistics table "Culture Descriptive Statistics" prove this average to be actually 15.79. The minimum ratio of number of cultures is found to be 1.6 (a value fitting between 0 and 10 in the histogram), and the maximum ratio of number of cultures is 60.5 (a value fitting between 60 and 70 in the histogram).

The density curve fits the distribution pretty well. Additionally with the right-skewed distribution, there are outliers located between the bounds 50 and 70. We can interpret this as some hospitals having conducting a very high number of cultures on patients without signs or symptoms of hospital-acquired infection.

Table 4: Culture Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
1.6	15.79	14.1	60.5	10.23

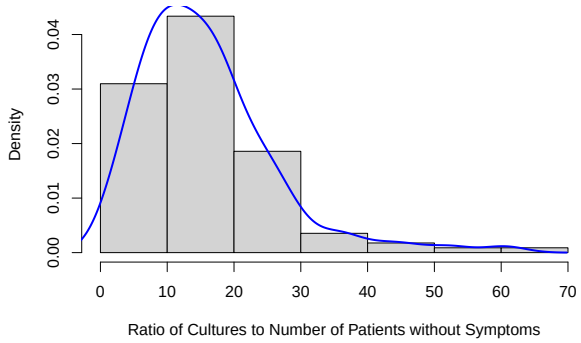


Figure 4: Histogram showing distribution of Culture Variable, with Density Curve Overlaid

3.2.2 Xray

The variable Xray measures the ratio of number of X-rays (taking an image of the body, particularly the chest, to look for conditions such as pneumonia) performed to number of patients without signs or symptoms of pneumonia, times 100.

The distribution of Xray is approximately bimodal as there are two distinct peaks: one between the bounds 80 and 90, and another between the bounds 50 and 60. This potentially indicates that there are two different groups in the dataset. For example, one group could be hospitals taking a lower number of Xrays for patients, and the other group could be taking a high number of Xrays on patients.

The descriptive statistics table "Xray Descriptive Statistics" reveals that the minimum ratio of number of X-rays performed to number of patients without symptons of pneumonia is 39.6, and the maximum ratio is 133.5. We find that the mean ratio is 81.63, and the median is 82.3.

The density curve fits the distribution pretty well.

Table 5: Xray Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
39.6	81.63	82.3	133.5	19.36

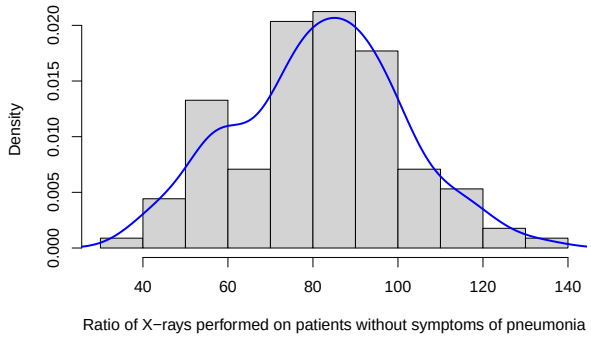


Figure 5: Histogram showing distribution of Xray Variable, with Density Curve Overlaid

3.2.3 Beds

The variable Beds measures the average number of beds in a hospital during study period.

Table 6: Beds Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
29	252.17	186	835	192.84

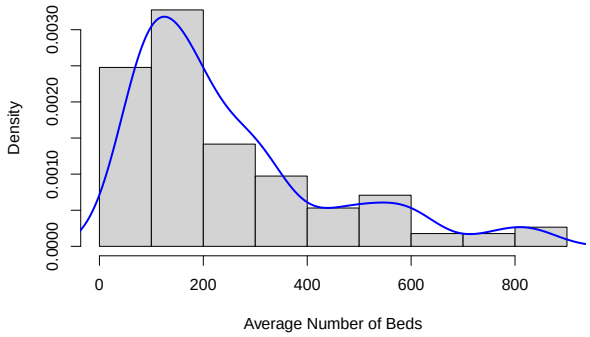


Figure 6: Histogram showing distribution of Beds Variable, with Density Curve Overlaid

The distribution of Beds from the histogram is skewed to the right, meaning there are outliers in the 600+ bed range. The minimum number of beds is 29 beds at one hospital, and the maximum number of beds is 835, which falls outside of our scale in the histogram. The mean number of beds is 252.17 beds, or about 252 beds, and the median number of beds is 186.

The density curve fits the distribution pretty well.

3.2.4 MedSchool

The variable MedSchool measures whether a hospital has an associated medical school, with 1 being Yes and 2 being No. We convert the values "1" and "2" into levels ("Yes" and "No") in section 2 "Data Description and Processing" for analysis.

From the table "Hospitals by Medical School Affiliation" and the bar chart "Bar Chart of MedSchool", we see that a majority of hospitals do not have an associated medical school, at 84.96%. There is a small minority of hospitals that have an associated medical school, at 15.04%.

The density curve fits the distribution pretty well.

Table 7: Hospitals by Medical School Affiliation

MedSchool	Count	Percent
Yes	17	15.04
No	96	84.96

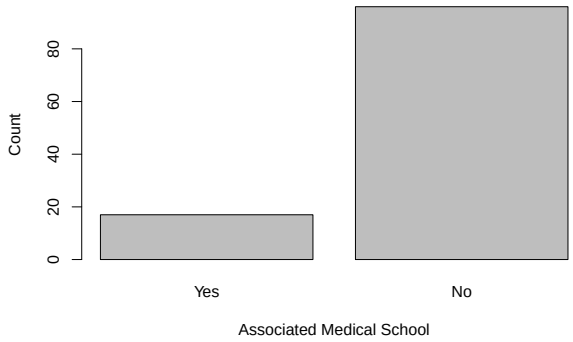


Figure 7: Bar Chart of the MedSchool Variable

3.2.5 Region

The variable Region measures the geographic region that a hospital is located in, where: 1 represents NE (Northeast), 2 represents NC (Northcentral), 3 represents S (South), and 4 represents W (West).

From the table "Hospitals by Geographic Region" and the bar chart "Bar Chart of Region", we see that most of the hospitals in our dataset are geographically located in the South, at 32.74%. Behind the South region is hospitals located in North Central, at 28.32%, followed by Northeast at 24.78%. The region with the least number of hospitals represented in our dataset is the West, at 14.16%.

The density curve fits the distribution pretty well.

Table 8: Hospitals by Geographic Region

Region	Count	Percent
Northeast	28	24.78
North Central	32	28.32
South	37	32.74
West	16	14.16

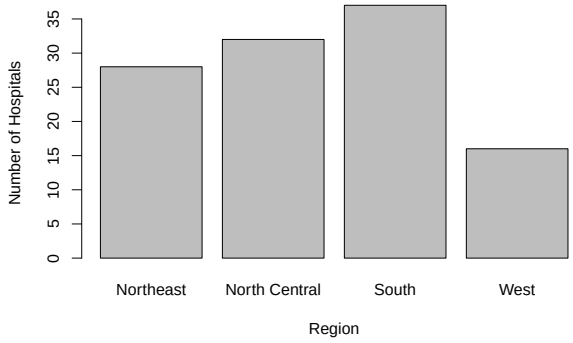


Figure 8: Bar Chart of the Region Variable, with Density Curve Overlaid

3.2.6 Census

The variable Census measures the average number of patients in hospital per day during the study period.

The distribution of Census is skewed to the right, with outliers between the bounds of 700 and 800 people in the histogram. The table "Census Descriptive Statistics" reveal that the minimum average number of people in a hospital is 20 people, the maximum average number of people in a hospital is 791, and the average number of people that visit a hospital is 191.37, or about 191 people. The median average number of people is 143.

The density curve fits the distribution pretty well.

Table 9: Census Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
20	191.37	143	791	153.76

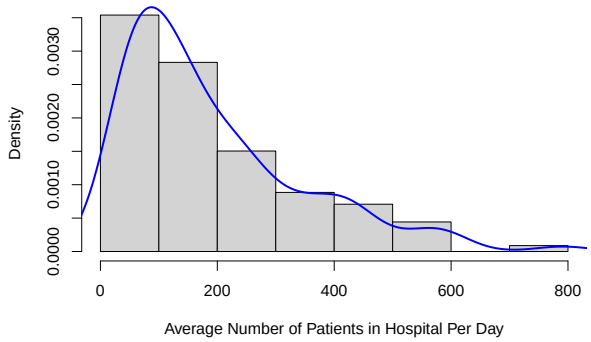


Figure 9: Histogram showing distribution of Census Variable, with Density Curve Overlaid

3.2.7 Nurses

The variable Nurses measures the average number of full-time equivalent registered and licensed practical nurses during study period. This number includes the number of full-time nurses plus one half the number of part-time nurses.

The distribution of Nurses is skewed to the right, with outliers between the bounds of 500 and 700 people in the histogram. A peak is found to be just below the 100 average number of nurses mark. The table "Nurses Descriptive Statistics" reveals that the minimum average number of nurses is 14, the maximum number is 656 nurses, and the mean average number of nurses is 173.25, or about 173 nurses. The median is 132 nurses. The right skewness of the histogram shows that there are few hospitals with an average number of Nurses over 500. Most of the hospitals have an average number of nurses in the range of 0 and 200, or about 173 based on the mean from the descriptive statistics.

The density curve fits the distribution pretty well.

Table 10: Nurses Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
14	173.25	132	656	139.27

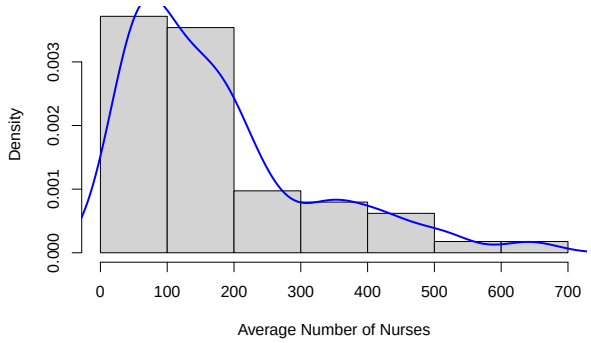


Figure 10: Histogram showing distribution of Nurses Variable, with Density Curve Overlaid

3.2.8 Facilities

The variable Facilities measures the percent of 35 potential facilities and services that are provided by the hospital.

The histogram of Facilities reveals a bimodal distribution, with two peaks: one peak between the bounds of 30 and 40, and another peak between the bounds of 50 and 60. This could mean that there could be two groups associated with this variable: a group of potentially rural hospitals with a lower percent of potential facilities and services, and another group of more wealthy/ middle class hospitals with a higher percent of 35 potential facilities and services at the hospital. The descriptive statistics reveal that the minimum percent of 35 potential facilities and services that are provided by the hospital is 5.7%, and the maximum percent is 80%. The mean percent is 43.16%, and the median percent is 42.9%.

The density curve fits the distribution pretty well.

Table 11: Facilities Variable Descriptive Statistics

Min	Mean	Median	Max	StdDev
5.7	43.16	42.9	80	15.2

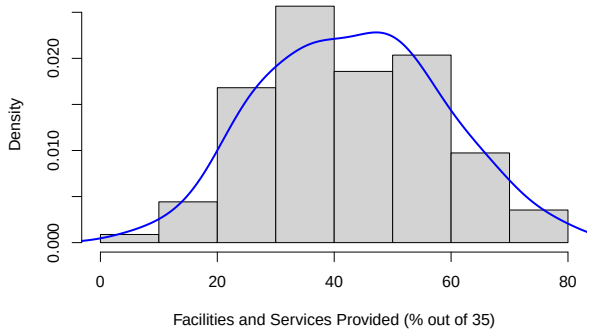


Figure 11: Histogram showing distribution of Facilities Variable, with Density Curve Overlaid

4 Modeling

For our models, infection risk is the dependent variable, and our ten predictors are: “Stay”, “Age”, “Culture”, “Xray”, “Beds”, “MedSchool”, “Region”, “Census”, “Nurses”, and “Facilities”.

4.1 Hypothesis

We hypothesize that hospitals with longer average patient stays have higher infection risk, after accounting for the other patient and hospital characteristics included in the model.

Our null hypothesis is that all predictor coefficients, excluding the intercept, equal zero. This means that the predictors, such as Beds or Age, have zero effect on the response variable, which is the average probability of acquiring an infection in the hospital (InfctRsk). Our alternative hypothesis is that at least one predictor coefficient is nonzero. This would mean that there is a predictor that has an effect on the average probability of acquiring an infection in the hospital, and it could be increasing or decreasing the probability of infection risk.

Using a significance level of 0.05, the overall F-test was statistically significant, with an F-statistic of 11.76 and a p-value of 1.963e-14. We find the p-value is less than the significance level, as 1.963e-14 is less than .05. Therefore, we reject the null hypothesis and conclude that at least one of our predictors has an effect on the average probability of hospital infection risk.

4.2 Model 1:

For our first model, we will include all ten predictors.

Holding other predictors constant, we find that from the table “Model 1 Fit Regression Coefficients & Significance for all Predictors”, the predictors Stay, Culture, Xray, Facilities, MedSchoolNo, and Region 1 (Northeast) to Region 4 (West) comparison were significant at the 0.05 level as each p-value was less than 0.05.

Note that Region1 (Northeast) is not in the table to prevent perfect multicollinearity. Instead, Region1 is a baseline to compare each region (i.e. Region2, Region3, and Region4) against it. For example, region 4’s values (e.g. coefficient of 1.15) is the expected difference in the outcome between Region4 and Region1. This is also seen with MedSchool, where MedSchoolYes (Hospitals with an associated medical school) is the baseline, and MedSchoolNo (Hospitals without an associated medical school) is compared against MedSchoolYes.

Table 12: Model 1 Fit Regression Coefficients & Significance for all Predictors

Predictor	Coefficient	Standard Error	P-value
(Intercept)	-2.51	1.32	0.06
Stay	0.24	0.07	0.00
Age	0.01	0.02	0.54
Culture	0.05	0.01	0.00
Xray	0.01	0.01	0.03
Beds	0.00	0.00	0.20
MedSchoolNo	0.66	0.32	0.04
Region2	0.43	0.26	0.11
Region3	0.37	0.27	0.18
Region4	1.15	0.34	0.00
Census	0.00	0.00	0.27
Nurses	0.00	0.00	0.29
Facilities	0.02	0.01	0.04

Each additional day increase (or one unit increase) in the variable Stay is associated with a 0.24 percent increase in infection risk. One unit increases in Culture is associated with an increase of approximately 0.054 percent in infection risk, as well as a one unit increase in Xray leads to an approximately .011553 percent increase of infection risk. A one percent point increase in Facilities was associated with an approximate 0.021 percentage-point infection risk increase.

Hospitals without medical school affiliations had infection risk 0.66 percentage points higher than affiliated hospitals. Hospitals in Region4 (West) had about 1.15 percentage points of higher infection risk than Hospitals in Region1 (Northeast).

Age, Beds, Census, Nurses, and the remaining regional comparisons were not significant. Culture had the smallest p-value, providing the strongest evidence of an association.

Table 13: Model 1 Overall Regression Model Fit Table

Measure	Value
R-squared (%)	58.54
Adjusted R-squared (%)	53.56
F-statistic	11.76
Model p-value	<0.01

The table “Model 1 Regression Model Fit Table” reveals that Model 1 explained 58.54% of the variation in infection risk across hospitals. After accounting for the predictor coefficients, the adjusted R-squared was 53.56%. The overall F-test was statistically significant (F-statistic is 11.76; p-value is less than 0.01 which is less than the chosen significance level of 0.05); at least one predictor was associated with infection risk.

The table “Model 1 Multicollinearity Assessment Table, showing VIF values for Predictors” reveals that the multicollinearity assessment identified high Variance Inflation Factor (VIF) values for Beds (35.74), Census (37.77), and Nurses (7.47). These values indicate substantial overlap between these predictors and other predictors in the model. All remaining values were below the threshold(5).

Table 14: Model 1 Multicollinearity Assessment Table

Predictor	VIF
Stay	2.41
Age	1.27
Culture	1.56
Xray	1.39
Beds	35.74
MedSchool	1.79
Region	1.83
Census	37.77
Nurses	7.47
Facilities	3.14

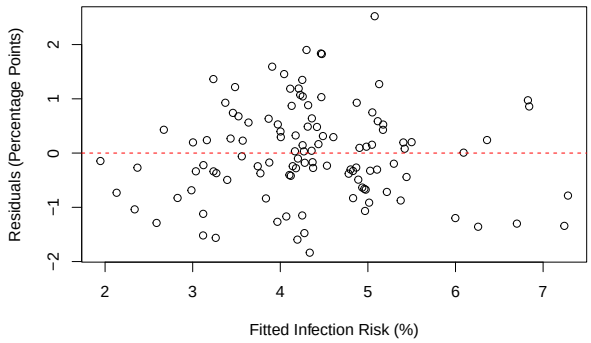


Figure 12: Model 1 Fitted vs. Residuals Plot

The figure "Model 1 Fitted vs. Residuals Plot" shows residuals as points distributed randomly on both sides of zero, without a pronounced funnel-shaped pattern. They are distributed above and below the red-dotted line at zero. A few large residuals are also present. The plot therefore suggests possible departures from a linear relationship. We don't see any trends in the residual points or in the red dotted line.

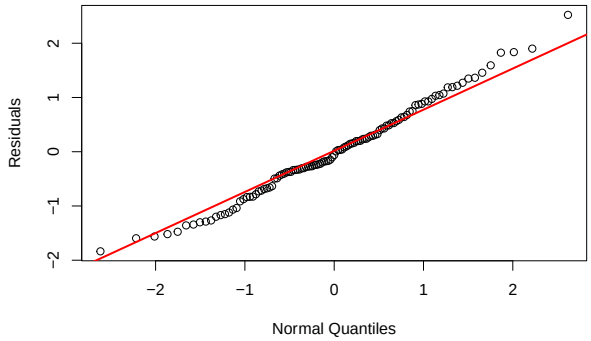


Figure 13: Model 1 Q-Q Plot

The figure "Model 1 Q-Q Plot" reveals that most residuals follow the red reference line closely. There are some points off the reference line at tails, especially where the largest positive residuals (Normal Quantiles greater than about 1.5+) lie above the line. This suggests residual normality, although the upper tail shows some deviation from the normal distribution.

Table 15: Model 1 Breusch-Pagan Test for Nonconstant Variance

Test Statistic	Degrees of Freedom	P-value
14.01	12	0.3

The "Model 1 Breusch-Pagan Test" table reveals that a conducted Breusch-Pagan Test produced a test statistic of 14.01 with 12 degrees of freedom and a p-value of 0.30. At the 0.05 significance level, we fail to reject (and thus accept) the null hypothesis of homoscedasticity (constant error variance). The test provides insufficient evidence of heteroscedasticity (nonconstant error variance), consistent with the absence of a funnel pattern in the residual plot.

4.3 Model 2:

For our second model, we will only include "Stay", "Culture", "Xray", "MedSchool", and "Region", and "Facilities". This model was chosen using stepwise selection.

Table 16: Model 2: Fit Regression Coefficients & Significance for Stepwise Selected Predictors

Predictor	Coefficient	Standard Error	P-value
(Intercept)	-1.94	0.83	0.02
Stay	0.27	0.06	0.00
Culture	0.05	0.01	0.00
Xray	0.01	0.01	0.03
MedSchoolNo	0.50	0.29	0.09
Region2	0.34	0.25	0.18
Region3	0.35	0.26	0.18
Region4	1.11	0.33	0.00
Facilities	0.03	0.01	0.00

From the table "Model 2: Fit Regression Coefficients & Significance for Stepwise Selected Predictors", if we holding the other included predictors constant, Stay, Culture, Xray, Facilities, and the Region1 (Northeast) to Region4 (West) comparison were significant at the 0.05 level.

Each additional day of average stay was associated with a 0.27 percent point increase in infection risk. One unit increases in Culture and Xray were associated with approximately 0.05 and 0.01 percent point increases, respectively. Each percentage-point increase in Facilities was associated with approximately 0.025 percentage points higher infection risk.

West hospitals (Hospitals in the West Region) had estimated infection risk of about 1.11 percent points for infection risk higher than Northeast hospitals (Hospitals in the Northeast Region). Medical school affiliation was not significant at the 0.05 level ($p = 0.09$), nor were the North Central to Northeast and South to Northeast region comparisons.

Table 17: Model 2 Overall Regression Model Fit Table

Measure	Value
R-squared (%)	57.12
Adjusted R-squared (%)	53.82
F-statistic	17.32
Model p-value	<0.01

From the "Model 2 Regression Model Fit Table", Model 2 reveals 57.12% of the variation in infection risk, with an adjusted R-squared of 53.82%. The overall F-test was statistically significant ($F = 17.32$, $p < 0.01$); at least one included predictor was associated with infection risk.

Table 18: Model 2 Multicollinearity Assessment Table

Predictor	VIF
Stay	1.67
Culture	1.36
Xray	1.37
MedSchool	1.48
Region	1.50
Facilities	1.48

The displayed VIF values in "Model 2 Multicollinearity Assessment Table, showing VIF values for Predictors" ranged from 1.36 to 1.67. Compared with Model 1, these results indicate reduced multicollinearity and less inflation of coefficient variances from overlapping predictors.

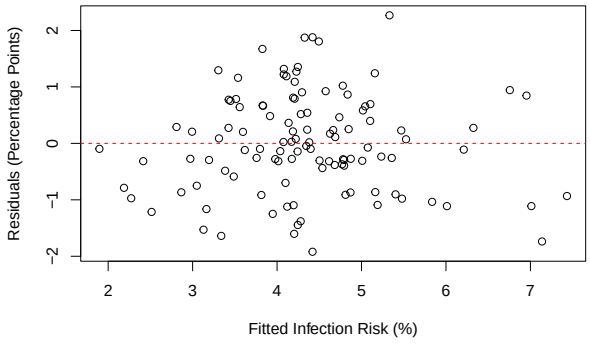


Figure 14: Model 2 Fitted vs. Residuals Plot

In "Model 2 Fitted vs. Residuals Plot", the residuals appear on both sides of zero. However, negative residuals at the lowest and highest fitted values suggest possible curvature. The reduced model therefore potentially departures from a linear relationship.

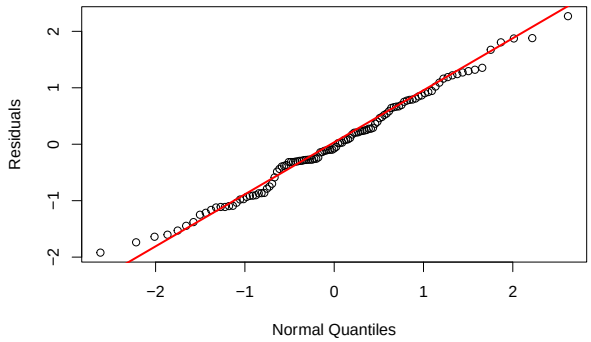


Figure 15: Model 2 Q-Q Plot

In “Model 2 Q-Q Plot”, the residuals generally follow the normal reference line, with minor departures in the tails. The plot supports approximate residual normality.

Table 19: Model 2 Breusch-Pagan Test for Nonconstant Variance

Test Statistic	Degrees of Freedom	P-value
6.83	8	0.56

The Breusch–Pagan test produced a statistic of 6.83 with 8 degrees of freedom and a p-value of 0.56. At the 0.05 significance level, we do not reject homoscedasticity (constant error variance). There is insufficient evidence of heteroscedasticity (nonconstant variance).

Model 2 retained six predictors rather than ten, while its adjusted R-squared increased slightly from 53.56% to 53.82%. It therefore achieved similar fit with fewer predictors and much less multicollinearity. # Predicting with Models 1 and 2

We evaluated both models using 10-fold cross-validation repeated three times. In each repetition, the data was divided into ten groups. Each model was trained using 90% of the observations and tested on the remaining 10%, until every group had been used for validation. Both models used the same splits, producing 30 validation results per model.

We measured prediction error using root mean squared error (RMSE), expressed in percentage points of infection risk. Lower RMSE indicates more accurate predictions. We summarized performance using the RMSE and its standard deviation across the validation samples.

Table 20: Model 2 Repeated Cross-Validation Prediction Performance

Model	RMSE	RMSE Standard Deviation
Model 1: Full	0.99	0.19
Model 2: Reduced	0.95	0.16

Model 1 had a mean RMSE of 99 percentage points, compared with 95 for Model 2. Their RMSE standard deviations were 0.19 and 0.16, respectively. Model 2 had the lower average prediction error and showed more consistent performance across validation samples.

5 Summary and Conclusions

Both regression models identified associations between patient and hospital characteristics and hospital-acquired infection risk. Model 1 included all ten predictors and explained 58.54% of the variation in infection risk, with an adjusted R-squared of 53.56%. Model 2 retained six predictors and explained 57.12% of the variation, with a slightly higher adjusted R-squared of 53.82%. Although their explanatory performance was similar, Model 2 had much less multicollinearity. The repeated cross-validation results also showed lower average prediction error and less variability in prediction error for Model 2. We therefore selected Model 2 as our preferred model.

Both models supported our hypothesis that longer average patient stays are associated with higher infection risk. In Model 2, each additional day of average stay was associated with a 0.27 percentage-point increase in infection risk, holding the other included predictors constant. Higher culture and X-ray testing ratios, higher availability of facilities and services, and being located in the West rather than the Northeast were also significantly associated with higher infection risk. These findings suggest that both patient and hospital characteristics help explain differences in infection risk across hospitals.

Several limitations affect these conclusions. The data is observational, so the associations do not establish cause and effect. Each observation represents a hospital, so the results cannot directly describe an individual patient's risk. Unmeasured factors may explain some of the observed relationships. Selecting the reduced model using the full dataset before cross-validation may make its estimated prediction performance optimistic.